

Fuelling The Need For Speed

Discover the technology that helps Karun Chandhok burn rubber on the British Formula 3 racing circuit

Meera Vankipuram

Boy wonder Karun Chandhok is, at 20, India's youngest auto racing sensation. Having made his debut in the JK Tyre Racing Championship three years ago, Chandhok is now the second Indian, after Narain Karthikeyan, to participate and notch up wins in the British Formula-3 circuit.

Chandhok, who began go-karting at the age of six, went on to win the national racing championship when he was still in school, and at 17, was the Formula Asia champion.

The British Formula 3 is regarded as the stepping-stone to the Formula 1 circuit—racing greats such as Ayrton Senna first made it big in F-3. The technology that fuels the sport is partly responsible for the exorbitant cost of the racing as well as the associated training. With an annual bill running into crores of rupees, Chandhok is heavily dependent on sponsors.

Chandhok and his team represent a perfect synergy of the Holy Trinity of the racing world—man, machine and technology.

The Nissan World Series car, which Karun moved to for the final four races of the 2004 racing season, is typically a single-seater powered by a three-litre V6 engine producing 450 BHP and capable of speeds of up to 290 kmph. Like in Formula 1, a podium finish is more a team effort than the prowess of the driver alone.

As Chandhok puts it, "The chassis of the car has been built by a company called Dallara after several long hours of CAD-CAM design and time in the wind tunnel. The engine is a 3000 cc V6 Nissan, built in the UK. It runs on software designed by Pi Research"—the software, Pi Delta, costs about Rs 15 lakh.

The data acquisition system developed by UK-based Pi Research, logs the various parameters of the car—speed, engine revs, suspension movement, steering movement, engine and oil temperature, and in-line and longitudinal g-forces.

The data acquisition room is a makeshift facility and is set up at every racetrack. "The team factory is in Italy but we travel all over Europe to different tracks—we have three systems hooked together on a land system, which functions as a



Speed racer Karun Chandhok seated in his Nissan World Series car

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data acquisition unit,” says Chandhok.

The logging is done with the help of 60 sensors along various parts of the car. “This information allows the engineers and the driver to understand the way the car behaves and evolve better training and racing strategies,” he adds.

Along with Chandhok, engineers Daniele and Roberto study the data and then figure out the best solutions to get the car to achieve its maximum efficiency and performance level.

Research, therefore, becomes an integral part of the training process. The research is usually aimed at attaining maximum fuel efficiency, better speeds and manoeuvring capabilities, and of course, more wins.

So what are the costs involved in keeping Chandhok's car in good humour? Explains Chandhok: “The costs work out to over Rs 2,000 per kilometre in the world series. It's hard to break down the maintenance costs but a set of tyres costs about Rs 40,000. A litre of fuel is Rs 110.”

For each season, Chandhok's racing budget is nearly Rs 3.2 crore—that's how much it takes to keep the driver, the car and the team in top condition. “My sponsors don't cover the entire cost, so at the moment, we have to cut back on testing and practice time to save costs,” he says.

Off the track, Chandhok uses his laptop to stay connected almost all the time. “I use my laptop extensively for e-mail and use the Net a lot nowadays to book flight tickets.” Chandhok owns a Compaq Presario 2500 series.

So what powers the Speed Racer? “Obviously, the thrill of winning—driving for the heck of it is a pointless exercise and a waste of time, effort and money!” Touché. ☒

meera_vankipuram@thinkdigit.com



Racing's Expensive

Chandhok races with the Nissan World Series car, the chassis of which was designed and built in the UK. The data acquisition software, Pi Delta, which helps Chandhok monitor his racing prowess, costs nearly Rs 15 lakh. In the World Series, racing costs work out to over Rs 2,000 a kilometre - a set of tyres, for example, costs about Rs 40,000. A litre of fuel is Rs 110. For each season, Chandhok's racing budget is nearly Rs 3.2 crore - a sum funded partially by sponsors.